

Adnan Harun Dogan

Personal Information

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Employment

Feb 2023–Present 	Research Assistant , Image Processing and Pattern Recognition Laboratory (ImageLab), METU Supervisor: Prof. Dr. Sinan Kalkan , Assoc. Prof. Dr. Emre Akbas <i>Ankara, Turkey</i>
Mar 2025–Oct 2025 	Graduate Intern , Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich Supervisor: Prof. Dr. Christian Holz <i>Zürich, Switzerland</i>
Oct 2021–Feb 2022 	Undergraduate Intern , Center for Artificial Intelligence and Robotics (ROMER), METU Supervisor: Assoc. Prof. Dr. Erol Sahin , Assoc. Prof. Dr. Bugra Koku <i>Ankara, Turkey</i>
Jun 2021–Sep 2021 	Undergraduate Intern , Cyber-Physical Systems Laboratory, University of California, Irvine Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque <i>Irvine, California, USA</i>

Education

2022–2025 	Master of Science in Computer Engineering , Middle East Technical University (METU) Advisor: Prof. Dr. Sinan Kalkan , Assoc. Prof. Dr. Emre Akbas Thesis: Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation <i>Ankara, Turkey</i>
2018–2022 	Bachelor of Science in Computer Engineering , Middle East Technical University (METU) <i>Ankara, Turkey</i>

Research Interests

My research centres on differentiable optimization for deep learning: how to propagate useful gradients through combinatorial and constrained optimization layers, and how to train networks end to end when the loss is ranking based or defined through optimal transport. I apply these ideas mainly to object detection. A second line of work is physics-informed and structured state estimation, in particular learned Kalman filters and conservation-aware neural surrogates for fluid dynamics, along with deep models for physiological signals such as EEG and inertial sensing.

Publications

Peer-reviewed publications

9. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. *International Conference on Robotics and Automation (IEEE ICRA)*. arXiv:2606.02251 (Code)
8. Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. *IEEE Transactions on Visualization and Computer Graphics*. doi.org/10.1109/TVCG.2025.3549132 (Code)
7. Amin, M. A., **Dogan, A. H.**, Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. *International Conference on Advanced Information Networking and Applications (AINA 2024), Lecture Notes on Data Engineering and Communications Technologies*, pp. 217-226. doi.org/10.1007/978-3-031-57942-4_22 (Data)
6. Yavuz, F., Cam, B. C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. *European Conference on Computer Vision (ECCV 2024), Lecture Notes in Computer Science*, vol 15117, pp. 93-109. doi.org/10.1007/978-3-031-73202-7_6 (arXiv, PDF, Code)
5. Sever, Y., & **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU Journal on Future and Evolving Technologies*, 4(2), 383-388. doi.org/10.52953/fplr8631 (PDF, Code)
4. Karagoz, P., Cekineli, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. In M. Akkar Ercan & K. Aydinler (Eds.), *Valorising Underground Built Heritage in Cappadocia* (pp. 61-92). CNR Edizioni. ISBN 978-88-8080-605-9. (PDF)
3. Sever, Y., Ekinici, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. *IEEE/SRMC'22, September 2022*. doi.org/10.1109/SRMC57347.2022.00007  **Best Paper Award**
2. Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. *Big Data Analytics and Knowledge Discovery (DaWaK 2021), Lecture Notes in Computer Science*, vol 12925, pp. 152-157. doi.org/10.1007/978-3-030-86534-4_14
1. Demirel, B. U., **Dogan, A. H.**, & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. *Computing in Cardiology (CinC)*. doi.org/10.23919/cinc53138.2021.9662935 (PDF, Code)

Preprints

2. **Dogan, A. H.**, Deniz, M., Alemdar, H., & Baran, O. U. (2026). CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws. arXiv preprint arXiv:2607.06587. arXiv:2607.06587
1. **Dogan, A. H.**, & Dogan, A. (2021). An assembled deep learning approach for flow field prediction. (PDF)

Manuscripts under review

3. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Deep Kalman state estimation for drift-calibrated full-body pose tracking from sparse inertial sensors. Under Review at Conference on Computer Vision and Pattern Recognition.
2. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Magnetometer-free state estimation for sparse inertial human pose tracking via motion-based yaw inference. Under Review at Conference on Computer Vision and Pattern Recognition.
1. **Dogan, A. H.**, Celtik, S. O., Baran, O. U., & Alemdar, H. (2026). Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows. Under Review at Journal of Aerospace Science and Technology.

Theses

1. **Dogan, A. H.** (2025). Incorporating piecewise linear functions with constant regions in backpropagation. M.Sc. Thesis, Middle East Technical University. (Thesis)

Research Experience

Sep 2021–Jun 2022



Cyber Security Research Group, METU

Undergraduate Research Project

Advanced research on container security with intrusion detection systems (CONTSEC) for cloud-native environments. Designed and implemented high-performance IDS pipeline optimized for containerized cloud infrastructures. Collaborated on Kubernetes misuse detection research, contributing to novel datasets and detection methodologies.

Advisor: Assoc. Prof. Dr. [Pelin Angin](#)

Ankara, Turkey

Jan 2020–Jun 2022



Image Processing and Pattern Recognition Laboratory (ImageLab), METU

Undergraduate Research Project

CNNFOIL: Approximating HLLC Riemann solver for flow prediction around airfoils via encoder-decoder neural networks. Engineered state-of-the-art neural network models achieving 4.4% performance improvement in predicting critical flow field parameters around airfoil geometries.

Advisor: Assist. Prof. Dr. [Hande Alemdar](#), Assist. Prof. Dr. [Baran Ugras](#)

Ankara, Turkey

Aug 2021–Oct 2021



TUBITAK BILGEM

Summer Intern

Cloud infrastructure development at Cloud Computing and Big Data Research Lab. Contributed to Sapphire Cloud platform enhancements.

Kocaeli, Turkey

Oct 2020–Oct 2021



Textual Event Graph HUB (TEGHub), METU

Undergraduate Research Project

Textual analysis for irony detection and research impact prediction. Implemented end-to-end neural pipelines integrating GCN and LSTM networks for analyzing citation patterns and textual content on Twitter datasets.

Advisor: Prof. Dr. [Pinar Karagoz](#)

Ankara, Turkey

Feb 2020–Jun 2021



School of Medicine, Koç University

Undergraduate Research Project

Healthcare policy simulation for evidence-based decision making. Utilized simulation models focusing on mechatronics applications in healthcare. Developed GCN-based citation graph embeddings for healthcare policy papers.

Advisor: Assist. Prof. Dr. [Ozge Karanfil](#)

Istanbul, Turkey

Aug 2020–Oct 2020



Enocta Inc.

Summer Intern

NLP and transformer model optimization. Engineered GPT-2 and T5 models using HuggingFace for enterprise learning management systems.

Ankara, Turkey

Teaching

Teaching Assistant, Computer Engineering Department, METU

Feb 2022–Present

8. CENG 280 - Automata Theory (Spring 2024, Spring 2025, Spring 2026)
7. CENG 242 - Programming Language Concepts (Spring 2022, Spring 2023, Spring 2025, Spring 2026)
6. CENG 223 - Discrete Computational Structures (Fall 2023, Fall 2024, Fall 2025)
5. CENG 424 - Logic for Computer Science (Fall 2023, Fall 2025)
4. CENG 460 - Introduction to Robotics for Computer Science (Fall 2024)
3. CENG 213 - Data Structures (Spring 2024)
2. CENG 382 - Analysis of Dynamical Systems (Spring 2023)

1. CENG 334 - Introduction to Operating Systems (Spring 2022)

Undergraduate Teaching Assistant, Computer Engineering Department, METU

Spring 2019–Fall 2021

3. CENG 240 - Programming with Python for Engineers (Fall 2021)
2. CENG 111 - Introduction to Computer Engineering Concepts (Fall 2020)
1. CENG 230 - Introduction to C Programming (Spring 2019)

Awards and Fellowships

2022–2023	TUBITAK 2247-C STAR’23 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2022	Best Paper Award by IEEE/SRMC 2022
2022	Graduated in the first quantile of the class by Middle East Technical University (METU)
2021–2022	TUBITAK 2247-C STAR’22 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2020–2021	TUBITAK 2247-C STAR’21 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2017	University Entrance Exam: ranked 1466th of 2.27M candidates by OSYM, Turkey

Computer Skills

Languages & Tools:	Python/PyTorch, C/C++, ROS2, Rust
DevOps & HPC:	GitHub, Docker, Slurm (Workload Manager), Apptainer
Differentiable Optimization:	Combinatorial Optimization, Optimal Transport, Deep State Space Models
Deep Learning:	Computer Vision, Self-Supervised Learning, Probabilistic Deep Learning, Deep Reinforcement Learning

Languages

Turkish:	Native
English:	Fluent (TOEFL iBT 109/120, 2022)